Environmental Sciences - Unit 01

Ecosystem

Detailed Graph-Based DDCET Study Notes

Syllabus Scope

Environment, ecosystem structure, biotic and abiotic components, food relationships, trophic levels, energy flow, ecological pyramids, natural cycles, ecosystem types and ecological balance.

Contents

1. Environment, ecology and ecosystem
2. Abiotic and biotic components
3. Producers, consumers and decomposers
4. Food chains and food webs
5. Trophic levels and energy flow
6. Ecological pyramids
7. Water, carbon and nitrogen cycles
8. Terrestrial ecosystems
9. Aquatic and artificial ecosystems
10. Ecological balance and human disturbance

Format

The notes use explanations, bullet points, concept maps, process diagrams and graphs. Gridded comparison tables are intentionally not used.

1. Environment, Ecology and Ecosystem

1.1 Environment

The environment is the total set of surroundings that influence an organism. It includes air, water, soil, light, temperature, minerals and every living organism that interacts with it. Environmental factors are connected: rainfall changes vegetation, vegetation changes soil moisture, and those changes affect animals and microorganisms.

Ecology is the study of relationships among organisms and between organisms and their physical surroundings. It asks questions such as why a species lives in one habitat, how food energy moves, and how disturbances change populations.

1.2 Levels of Ecological Organisation



A habitat is the place where an organism lives. A niche is its functional role — how it obtains food, uses space, reproduces and interacts with other organisms. Several populations form a community, and the community together with non-living factors forms an ecosystem.

1.3 Ecosystem as a Functional Unit

An ecosystem is recognised by processes, not only by boundaries. Producers capture energy, consumers transfer it through feeding, decomposers return nutrients, and abiotic factors regulate every stage. A puddle, pond, crop field, forest and ocean can all be studied as ecosystems.

> **DDCET Focus**
>
> An ecosystem always includes both biotic and abiotic components. The sea/ocean is the largest ecosystem on Earth.

2. Abiotic and Biotic Components

2.1 Abiotic Components

Abiotic components are non-living physical and chemical factors. Sunlight supplies energy for photosynthesis. Temperature controls enzyme activity and species distribution. Water is required for metabolism and transport. Soil provides anchorage, minerals and a habitat for countless organisms. Air supplies oxygen, carbon dioxide and nitrogen.

Rainfall, humidity, wind, salinity and pH influence the type of ecosystem that develops. A small change in pH can change nutrient availability; a change in salinity can exclude freshwater organisms.

2.2 Biotic Components



Producers are autotrophs such as green plants and algae. Primary consumers feed on producers. Secondary and tertiary consumers feed on other consumers. Omnivores use plant and animal food. Scavengers eat dead animals, detritivores break dead matter into smaller pieces, and decomposers chemically break it down.

2.3 Interdependence

A plant needs light, water, carbon dioxide and minerals. A herbivore depends on plants; a carnivore depends on prey; decomposers depend on dead remains. When decomposition releases inorganic nutrients, plants can absorb them again. This interaction keeps matter moving through the ecosystem.

Canopy — tall trees, birds and sunlight
Understorey — shrubs and young trees
Forest floor — litter, fungi and decomposers
Soil — roots, minerals, water and microbes

A simplified forest vertical profile

3. Producers, Consumers and Decomposers

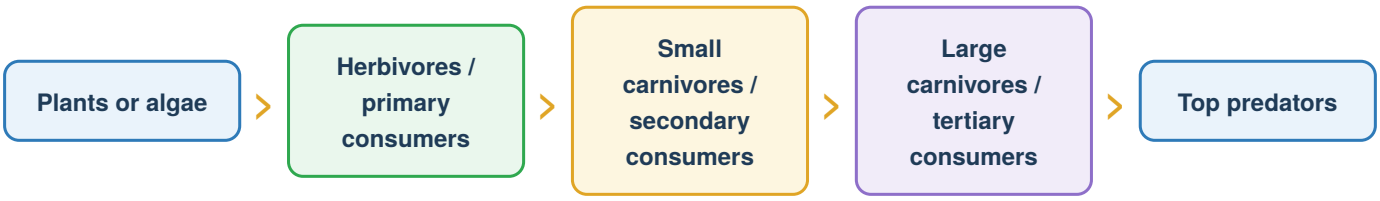
3.1 Producers and Productivity

Producers convert light energy into chemical energy stored in organic matter. Gross primary productivity is the total energy captured by producers. Producers use some of it in respiration; the remainder is net primary productivity, which supports growth and becomes available to consumers.

The exact productivity of an ecosystem depends on light, water, temperature, nutrients and the length of the growing season. Tropical forests can have high productivity, whereas deserts are strongly limited by water.

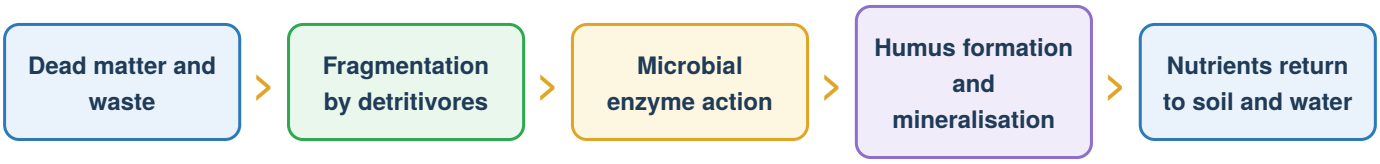
> **Supporting Concept**
>
> Net primary productivity = total energy captured − energy used by producers in respiration. DDCET usually tests the idea rather than calculation.

3.2 Consumer Levels



A consumer may occupy more than one trophic level. A human eating rice acts as a primary consumer; the same person eating fish may act at a higher level. Trophic level describes the feeding position in a particular pathway, not a permanent label for every organism.

3.3 Decomposition



Warmth, moisture and oxygen usually speed decomposition. Cold, dry or oxygen-poor conditions slow it. Energy stored in dead matter is eventually released as heat, while mineral nutrients become available for new growth.

4. Food Chains and Food Webs

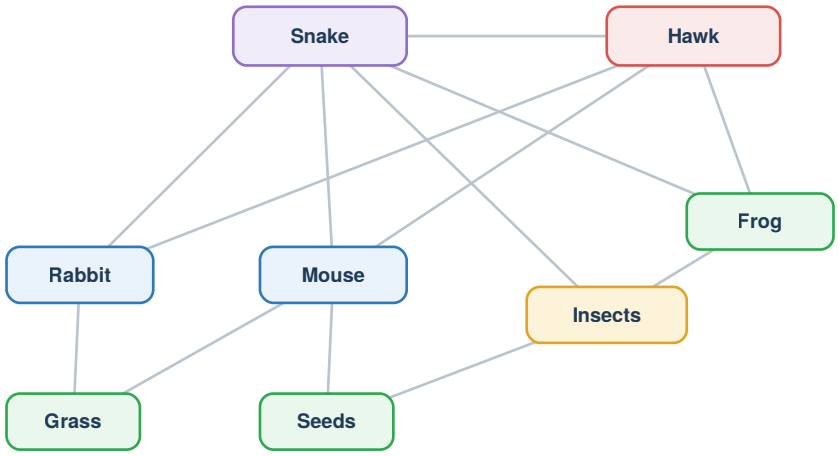
4.1 Food Chain

A food chain is a linear feeding sequence. The arrow shows the direction in which energy passes. In grass > grasshopper > frog > snake > eagle, the grasshopper receives energy from grass and the frog receives energy from the grasshopper.

A grazing food chain begins with a living producer. A detritus food chain begins with dead organic matter, such as leaf litter, and continues through detritivores and predators.

4.2 Food Web

Real ecosystems contain interconnected food chains because organisms usually have several food sources and predators. A food web provides alternate energy pathways and can increase stability. However, removal of an important producer or top predator may still disturb many links.



Lines show feeding links between organisms in the web.

4.3 Why Arrows Matter

In ecological diagrams, arrows represent energy transfer, not the action of eating. If a rabbit eats grass, the arrow points from grass to rabbit. Misreading the arrow is a common examination error.

> **Practice Idea**
>
> Trace what may happen if insects decline: frogs lose food, snakes may lose prey, and plants may experience less herbivory. A food-web question often asks for indirect effects.

5. Trophic Levels and Energy Flow

5.1 One-Way Flow of Energy

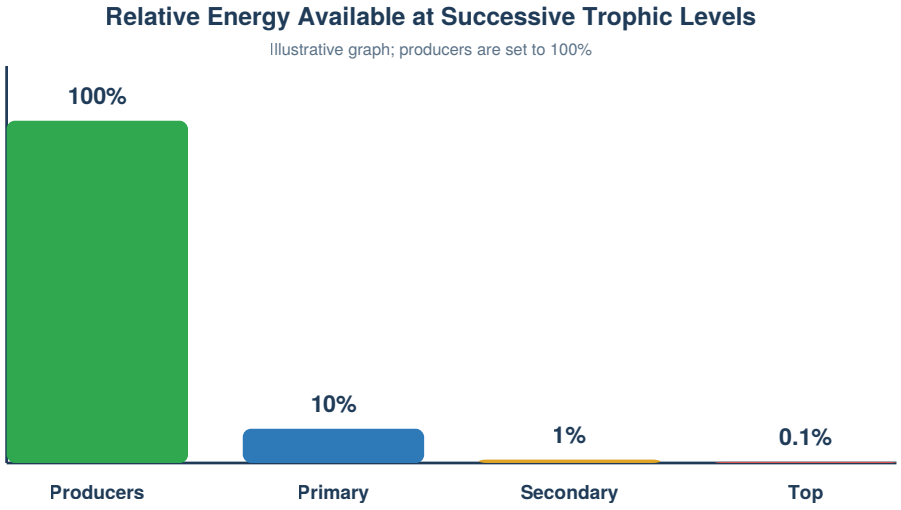
Energy enters most ecosystems as sunlight. Producers store only a small fraction as chemical energy. Organisms use energy for movement, repair, growth and respiration, and much of it leaves as heat. Heat is not recycled back into sunlight, so ecosystem energy flow is unidirectional.

5.2 Ten-Percent Rule

As a school-level approximation, about ten percent of the energy available at one trophic level reaches the next. The actual value varies, but the rule explains why higher levels contain less energy, why top predators are fewer, and why food chains are usually short.

Producers — about 10,000 energy units
Primary consumers — about 1,000 units
Secondary consumers — about 100 units
Top consumers — about 10 units

Conceptual application of the 10% transfer rule (widest = producers)



5.3 Energy Flow vs Nutrient Cycling

Energy moves through and leaves the ecosystem. Nutrients such as carbon, nitrogen and phosphorus circulate repeatedly between organisms and the environment. Decomposers are central to nutrient recycling but cannot recycle energy.

6. Ecological Pyramids

6.1 Pyramid of Numbers

A pyramid of numbers shows the count of organisms at each trophic level. It may be upright, as in grassland with many grasses and fewer herbivores, or inverted, as when one tree supports many insects.

6.2 Pyramid of Biomass

A biomass pyramid shows the total mass of living material. It is often upright on land. In some aquatic systems, the standing biomass of phytoplankton may be lower than that of consumers because phytoplankton reproduce rapidly and are consumed quickly.

6.3 Pyramid of Energy

The energy pyramid is always upright because energy is lost as heat at every trophic transfer. It gives the clearest picture of ecosystem energy flow and avoids the shape reversals possible in number or biomass pyramids.

Producers — about 10,000 energy units
Primary consumers — about 1,000 units
Secondary consumers — about 100 units
Top consumers — about 10 units

Conceptual application of the 10% transfer rule

> **High-Yield Point**
> The energy pyramid cannot be inverted. Number and biomass pyramids may be inverted in special ecosystems.

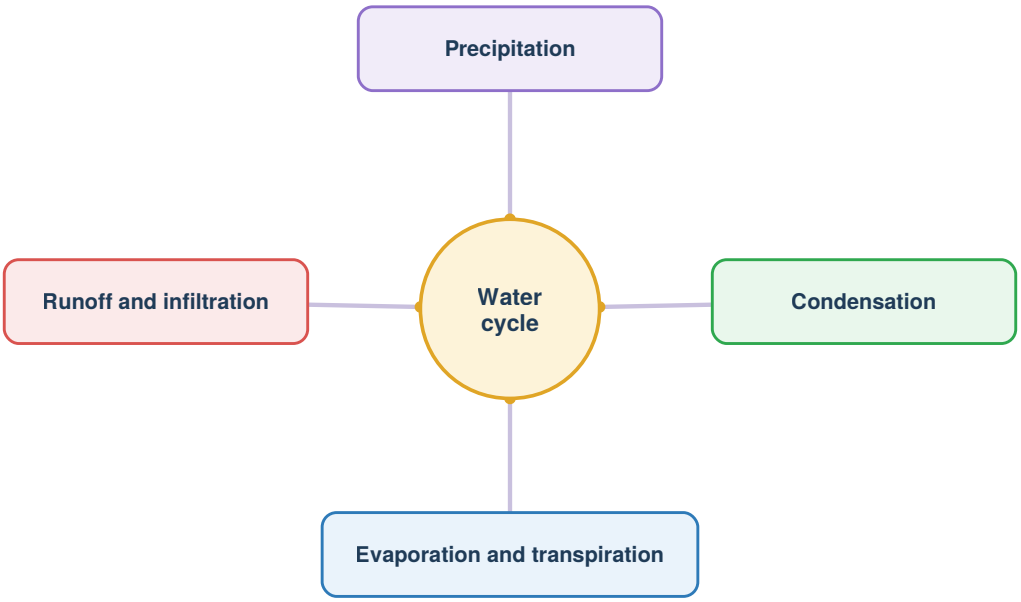
6.4 Ecological Efficiency

Ecological efficiency is the fraction of energy transferred to the next level. It depends on digestibility, metabolic rate, temperature and feeding behaviour. The ten-percent value is a convenient approximation, not a fixed law for every ecosystem.

7. Natural Cycles

7.1 Water Cycle

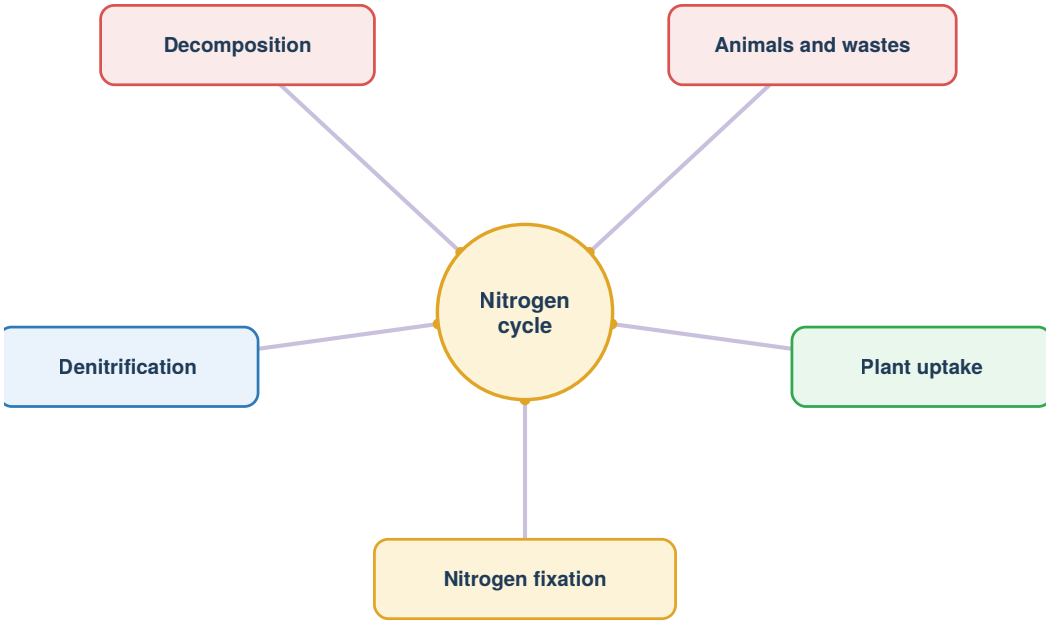
Solar energy drives evaporation from water bodies and transpiration from plants. Water vapour condenses into clouds, returns as precipitation, and moves through runoff, infiltration and groundwater. The cycle supplies freshwater and connects atmosphere, land and living organisms.



7.2 Carbon and Oxygen Cycles

Photosynthesis removes carbon dioxide and releases oxygen. Respiration, decomposition and combustion consume oxygen and return carbon dioxide. Oceans, vegetation and soils store carbon for different lengths of time. Deforestation reduces carbon storage and fossil-fuel burning adds old stored carbon to the atmosphere.

7.3 Nitrogen Cycle



Most organisms cannot use atmospheric nitrogen directly. Nitrogen-fixing microorganisms convert it into compounds plants can absorb. Animals obtain nitrogen through food. Decomposition returns compounds to soil, and denitrifying microbes return nitrogen to the atmosphere.

8. Terrestrial Ecosystems

8.1 Forest Ecosystem

Forests have layered vegetation, high biomass and complex food webs. Trees dominate production, leaf litter supports decomposers, and vegetation protects soil and regulates water flow. Deforestation removes habitat, releases carbon and increases erosion.



A simplified forest vertical profile

8.2 Grassland Ecosystem

Grasslands are dominated by grasses and herbs. Seasonal rainfall, grazing and fire strongly influence them. Extensive roots stabilise soil and store carbon. Overgrazing can remove protective cover and cause erosion or desertification.

8.3 Desert Ecosystem

Deserts have very low rainfall and large daily temperature variation. Plants reduce water loss through spines, waxy surfaces, deep roots or water-storage tissues. Many animals are nocturnal, burrow underground or produce concentrated urine to conserve water.

8.4 Comparing by Limiting Factor

Forests are commonly limited by light beneath the canopy or by nutrients. Grasslands are strongly shaped by seasonal water and grazing. Deserts are mainly limited by water. The same ecological principles operate, but the dominant limiting factor changes.

9. Aquatic and Artificial Ecosystems

9.1 Freshwater Systems

Ponds and lakes contain standing water; rivers contain flowing water. Light, depth, temperature, current and dissolved oxygen influence organisms. Wetlands filter some pollutants, store floodwater and provide breeding habitat. Excess nutrients can disturb the oxygen balance through eutrophication.

9.2 Marine Systems

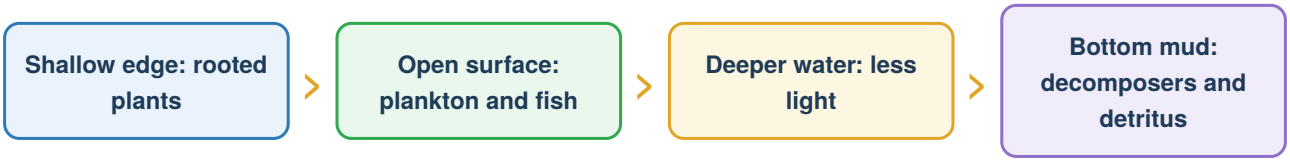
Marine ecosystems include oceans, seas, coasts, coral reefs and estuaries. They contain saline water and cover most of Earth. Phytoplankton are important producers. Ocean currents transport heat, nutrients and organisms across great distances.

> **DDCET Fact**
> The sea/ocean is the largest ecosystem on Earth.

9.3 Artificial Systems

Crop fields, aquariums, reservoirs and urban parks are artificial ecosystems because humans manage many inputs. They still contain producers, consumers, decomposers and abiotic factors, but irrigation, feeding, fertiliser or cleaning may be necessary to maintain the desired condition.

9.4 Pond Zonation



10. Ecological Balance and Human Disturbance

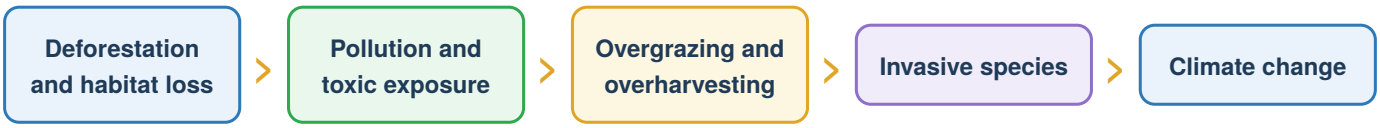
10.1 Dynamic Balance

Ecological balance does not mean that populations never change. Seasons, migration, reproduction and natural disturbances create variation. Balance means that key processes and populations remain within ranges that allow the ecosystem to continue functioning.

10.2 Biodiversity and Resilience

Biodiversity can increase resilience because several species may perform related functions. When one declines, another may partly maintain pollination, decomposition or food supply. Loss of a keystone species, however, may cause large changes.

10.3 Major Human Disturbances



Deforestation removes producers and habitat. Overgrazing exposes soil. Pollution changes air, water and food chains. Invasive species may outcompete native organisms. Climate change alters temperature, rainfall and seasonal timing. These disturbances can act together.

10.4 Protection

- Protect and restore forests, wetlands and riverbanks.
- Use soil, water, fisheries and forests at rates that allow recovery.
- Reduce pollution at the source and treat waste before release.
- Maintain wildlife corridors and control invasive species.
- Conserve biodiversity inside and outside protected areas.

Unit Outcome

You should be able to identify ecosystem components, trace food and energy relationships, explain nutrient cycling, compare ecosystem types and predict the effects of disturbances.